

NE 5350 – Nuclear Criticality Safety Project #1

Gabriel Lewis

University of Idaho • Idaho Falls Center for Higher Education
Nuclear Engineering and Industrial Management Department

galewis99@outlook.com

2026.09.24

1 Install OpenMC

The following figure depicts the installed OpenMC version and ENDF/B-VII.1 library.

```
(openmc-env) [galriel@archlinux ~]$ openmc --version
OpenMC version 0.15.3
Commit hash: 27e38e894697bb32a1dac7848d2618818b6b8daf
Copyright (c) 2011-2025 MIT, UChicago Argonne LLC, and contributors
MIT/X license at <https://docs.openmc.org/en/latest/license.html>
Build type: Release
Compiler ID: GNU 14.3.0
MPI enabled: yes
Parallel HDF5 enabled: yes
PNG support: yes
DAGMC support: yes
LibMesh support: no
MCPL support: no
Coverage testing: no
Profiling flags: no
JWUW support: no
(openmc-env) [galriel@archlinux ~]$
```

Figure 1. Screenshot of Terminal Depicting OpenMC Version

```
(openmc-env) [galriel@archlinux opt]$ cd nuclear_data
(openmc-env) [galriel@archlinux nuclear_data]$ ls
endfb71.tar.xz endfb-vii.1-hdf5
(openmc-env) [galriel@archlinux nuclear_data]$ cd endfb-vii.1-hdf5/
(openmc-env) [galriel@archlinux endfb-vii.1-hdf5]$ ls
cross_sections.xml neutron photon wmp
(openmc-env) [galriel@archlinux endfb-vii.1-hdf5]$ echo cross_sections.xml
cross_sections.xml
(openmc-env) [galriel@archlinux endfb-vii.1-hdf5]$ vim cross_sections.xml
(openmc-env) [galriel@archlinux endfb-vii.1-hdf5]$
::1 localhost
(openmc-env) [galriel@archlinux endfb-vii.1-hdf5]$
```

Figure 2. Screenshot of ENDF/B-VII.1 Folder

2 Compute the Critical Enrichment of Uranium

Create an infinite-homogeneous composition of pure uranium (metal). Then, use the enrichment argument to specify the uranium enrichment. For example, `uranium.add-element("U", 1.0, enrichment=4.0)` would specify a material with 4 % enriched (by mass) uranium.

As described in the `slab.py` example, an infinite-homogeneous composition of pure uranium metal can be represented in an infinite slab (in the y-direction) with a width of 6.5 cm. Below is the written code to express what is written in the prompt. Be aware that I have setup my location for the cross-sections.xml file differently than what is provided in the example and lecture. [See attached code here.](#)

- **Perform a criticality search to find the enrichment of uranium metal that would be critical in an infinite-homogeneous mixture.**

In an infinite slab with a width of 6.5 cm, the critical enrichment in the homogeneous mixture is approximately 5.9 %. [See attached code here for Question 2a.](#)

- **Without performing any additional calculations, using only your answer to 2(a), what would be the critical radius of a sphere of 3.2 % enriched uranium? Explain your reasoning.**

The critical radius for a sphere enriched at 3.2 % is unattainable. The infinite-homogeneous

mixture, without any reflector material is critical at 5.9 % with a width of 6.5 cm. How can it be, for a finite sphere with less enrichment, to be critical in the absence of any reflector material? Even the demon cores, composed of plutonium, needed tungsten carbide or beryllium, to become prompt supercritical.

Consider the six factor formula[1],

$$k_{eff} = \eta f p \epsilon P_{FNL} P_{TNL} = k_{\infty} P_{FNL} P_{TNL}$$

where η is the thermal fission factor, f is the thermal utilization factor, p is the resonance escape factor, ϵ is the fast fission factor, P_{FNL} is the fast non-leakage probability, and P_{TNL} is the thermal non-leakage probability. Since the product of the non-leak probabilities, P_{NL} is less than or equal to one, being a probability, for any finite geometry, k_{eff} can never exceed k_{∞} . Therefore, k_{eff} is only critical if k_{∞} is more than one, which can be viable provided enough enrichment of U-235.

2.1 Calculate the critical dimensions of High-Assay Low-Enriched Uranium (HALEU)

Low-Enriched Uranium (LEU) is defined by the International Atomic Energy Agency (IAEA) as less than 20 % enriched uranium. Historically in the United States, enrichment has been limited to less than 5 % as an administrative control. Recent interest in advanced reactors, especially fast-spectrum reactors like Sodium-cooled Fast Reactors (SFRs), has turned to HALEU to achieve the uranium density requirements for these reactors. Generally speaking, HALEU is defined as enrichments between 5 % and 20 %.

You will compute the critical dimensions of HALEU for certain geometries.

- Compute the critical radius for a sphere of 20 % enriched uranium.

The critical radius for a sphere of 20 % enriched uranium, in a vacuum, is approximately 1.025 cm. [See attached code here.](#) Be mindful of the `cross-section.xml` routing if you want to run it yourself.

- Compute the critical radius for a one-dimensional cylinder (infinite in the z-direction) of 20 % enriched uranium.

The critical radius for a one-dimensional cylinder, infinite in the z-direction, enriched to 20 % U-235 is approximately 0.76 cm, more so 0.759 cm. [See attached code here.](#)

- Compute the critical width of a one-dimensional slab of 20 % enriched uranium.
The critical width for this 20 % enriched slab is approximately 0.925 cm. [See attached code here.](#)
- Consider your answer to Problem 2 (above). How would a facility that is currently designed to handle 5 % enriched uranium need to be modified to accept 20 % enriched uranium? Recall that criticality safety is always a balancing act. What are the competing forces in this proposed design change?

In order for a facility, adapted for use for 5 % enriched uranium, to adapt to handle higher enriched uranium – up to 20% – changes to the relevant procedures will need to be made in order to maintain criticality safety. Such changes could include the dimensions for favorable geometry for storing the higher-enriched material and ensuring proper separation

between samples. The two competing forces in the design change for the facility is to balance Criticality Safety and Profitability. Having zero uranium or plutonium on site would be too extreme; what would be the purpose of the facility? Having no regard for criticality control, for increased business and profit margins, would result in bodily harm, or even death, for workers on site. To effectively make the changes to facility, a balanced approach is necessary to uphold criticality safety and economic viability.

References

- [1] Duderstadt, James J., Hamilton, Louis J., 1976. Nuclear Reactor Analysis. John Wiley & Sons, New York. ISBN 0471223638.